Math 5050 – Special Topics: Tensor Analysis– Spring 2026
w/Professor Berchenko-Kogan

Notes and Observations

Semester Plan

One chapter per week with the hopes of all 17 chapters by the end of April.

Status:

January 20, 2026 Reading Chapter 8 and Chapter 7 exercises.

Important Quotes

"A tensor is not meaningful by itself ... tensors lead to invariants." – Page 86

Indices:

The upper index is for covariant tensors and the lower index is for contravariant tensors. Notice that

$$\frac{\partial f^i}{\partial x^j}$$

that f is a function and x is a vector. This is in keeping with the definition from *Manifolds* where a covariant tensor is a functional and a contravariant tensor is vector. In the special case of the dual space then we know that a basis vector is of the form $f^i(x_j) = \delta^i_j$ which derives like a constant. In less precise terms, a functional is made up of terms based on the coordinates. Extending that notion, a tensor is rearrangement of coordinates (and applied in an expression) that responds to a change of coordinates in a specific way.

Progression of Intrinsic Objects

- Start with scalar field.
- Given a Position Vector $\mathbf{R}(Z^1, Z^2, Z^3)$ (no coordinates).
- Default tangent space $\mathbf{Z}_i = \frac{\partial \mathbf{R}}{\partial Z^i}$ also called covariant tensors.
  1. **Vector notation**

$$\mathbf{V} = \sum_{i=1}^{n} V^i \mathbf{Z}_i = V^i \mathbf{Z}_i$$

     It seems a little silly but this $V^i\mathbf{Z}_i$ is a new intuitive form of a vector (the index is arbitrary and unique).
- Covariant metric tensor $Z_{ij} = \mathbf{Z}_i \cdot \mathbf{Z}_j$. The foundational object of Reimannian Geometry.
- Contravariant metric tensor $Z^{ij}Z_{ij} = \mathbb{I}$ (inverse of the covariant metric tensor).
- Contravariant tensor derived from contravariant metric tensor, $\mathbf{Z}^j = Z^{ij}\mathbf{Z}_i$

Useful identitites

$$Z_{ij}Z^{ij} = \mathbb{I}$$
$$Z_{ij}Z^{ik} = \delta^k_j$$

a little dissection*

$$
\begin{aligned}
L &= \int_a^b \sqrt{Z_{ij}\frac{dZ^i}{dt}\frac{dZ^j}{dt}}\,dt \\
&= \int_a^b \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} Z_{ij}\frac{dZ^i}{dt}\frac{dZ^j}{dt}}\,dt \\
&= \int_a^b \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} (\,\mathbf{Z}_i\cdot\mathbf{Z}_j\,)\frac{dZ^i}{dt}\frac{dZ^j}{dt}}\,dt && \mathbf{Z}_i \perp \mathbf{Z}_j, i \neq j \\
&= \int_a^b \sqrt{\sum_{i=1}^{n}\left(\frac{dZ^i}{dt}\right)^2}\,dt
\end{aligned}
$$

Christoffel Symbols

By definitions:

$$
\begin{aligned}
\Gamma^k_{ij} &= \mathbf{Z}^k\cdot\frac{\partial\mathbf{Z}_i}{\partial Z^j} && (5.60)\\
&= -\frac{\partial\mathbf{Z}^k}{\partial Z^j}\cdot\mathbf{Z}_i && (5.64)
\end{aligned}
$$

Note: this is NOT a contraction. Picture three 3×3 matrices with these scalars. Each element is the projection of a partial derivative of a basis vector (i with respect to j) onto another basis vector, k. The collection of the elements of k make up the coordinates of the partial i with respect to j.

The contravariant version is

$$\frac{\partial\mathbf{Z}^k}{\partial Z^j} = -\Gamma^k_{ij}\mathbf{Z}^i \qquad (5.65)$$

note the contraction on i.

And based on the metric tensors

$$\Gamma^k_{ij} = \frac{1}{2}Z^{km}\left(\frac{\partial Z_{mi}}{\partial Z_j} + \frac{\partial Z_{mj}}{\partial Z_i} - \frac{\partial Z_{ij}}{\partial Z_m}\right) \qquad (5.67)$$

Transformation of Coordinates

$$\Gamma^i_{jk} = \overline{\Gamma}^u_{rs}\frac{\partial x^i}{\partial\overline{x}^u}\frac{\partial\overline{x}^r}{\partial x^j}\frac{\partial\overline{x}^s}{\partial x^k} + \frac{\partial^2\overline{x}^r}{\partial x^j\partial x^k}\frac{\partial x^i}{\partial\overline{x}^r}$$

Multiple change of coordinates

I find the notation for the Jacobian misleading at times. We must understand that J^i_j does not mean indices of i and j but different change of coordinate systems. Further the over use of Z^j when we might be meaning an entirely different basis vector gets confusing as well. Suppose a represents Cartesian Coordinates (x, y) and b represents polar coordinates (r, θ) then

$$J^a_b = \begin{pmatrix} x_r & x_\theta \\ y_r & y_\theta \end{pmatrix}$$

using the susbscript partial derivative notation. Now, let c represent an affine coordinate system p, q then

$$J^b_c = \begin{pmatrix} r_p & r_q \\ \theta_p & \theta_q \end{pmatrix}$$

Notice that combining these two we get

$$\begin{aligned} J^a_b J^b_c &= \begin{pmatrix} x_r & x_\theta \\ y_r & y_\theta \end{pmatrix} \begin{pmatrix} r_p & r_q \\ \theta_p & \theta_q \end{pmatrix} \\ &= \begin{pmatrix} x_r r_p + x_\theta \theta_p & x_r r_q + x_r \theta_q \\ y_r r_p + y_\theta \theta_p & y_r r_q + y_r \theta_q \end{pmatrix} \end{aligned}$$

Notice that $x_r r_p = \frac{\partial x}{\partial r}\frac{\partial x}{\partial p} = \frac{\partial x}{\partial p}$. Thus,

$$J^a_b J^b_c = \begin{pmatrix} x_p + x_p & x_q + x_q \\ y_p + y + p & y_q + y_q \end{pmatrix} = 2J^a_c$$

the b in the superscript followed by the subscript of J effectively cancel out, but b is not an index. Remember that it represents a change of coordinates that has become superfluous.

On Page 76 We have the identities (4.74) and (4.76)

$$J^i_{i'} J^{i'}_j = \delta^i_j \tag{4.74}$$
$$J^{i'}_i J^i_{j'} = \delta^{i'}_{j'} \tag{4.76}$$

Abuse of notation. From above we have $J^a_b J^b_c = J^a_c$ where each J is a distinct coordinate transformation between three different systems. Here, though, we are actually saying that i, j and i', j' are the same coordinate system but with different indices. We can see the contradiction in J^a_c. This implies $\delta^a_c = 1$ if $x = p$ (which might be true in the affine case). Consider in stead

$$\begin{aligned} J^b_a &= \begin{pmatrix} r_x & r_y \\ \theta_x & \theta_y \end{pmatrix} \\ J^b_a J^a_c &= \begin{pmatrix} r_x x_p + r_y y_p & r_x x_q + r_y y_q \\ \theta_x x_p + \theta_y y_p & \theta_x x_q + \theta_y y_q \end{pmatrix} \\ &= 2 \begin{pmatrix} r_p & r_q \\ \theta_p & \theta_q \end{pmatrix} \end{aligned}$$

which is distinctly NOT δ^b_c.

On page 80 we see this abuse again where i, j and i', j' are considered different indices of the same coordinate transformation.

Beware the metric tensors. Notice that the covariant metric tensor is defined as

$$Z_{ij} = \mathbf{Z}_i \cdot \mathbf{Z}_j.$$

This only makes sense when i, j are indices of the same coordinate system. The notation, though, suggests that they could be from different coordinate systems. Which might look something like

$$Z_{y\theta} = y \cdot \theta \tag{?}$$

Indeed, later expressions treat them that way as in

$$Z_{ij} J^i_{i'} J^j_{j'} = Z_{i'j'}$$

In this case, $J^i_{i'}$ and $J^j_{j'}$ are the same matrix (and must be the same) with different indices.

Page 85 we say "S^i_j is a tensor then S^i_i is an invariant." In this case, can be different coordinate systems. Let i be Cartesian and j be polar coordinates. J^j_i would be a change of coordinates from polar to Cartesian which makes $\bar{S}^i_i = S^i_j J^j_i$ which would make sense but $\bar{S}$ is not made up of the same coordinate systems as S. Be wary of S^i_i when the subscript is originally from a different coordinate system. The S refers to the tensor object S but S^i_i and S^i_j are very different flavors of it. This can be deceptive.

Covariant Derivative ∇_j

- Contravariant Tensor

$$\nabla_j V^i = \frac{\partial V^i}{\partial Z^j} + \Gamma^i_{jm} V^m$$

- Covariant Tensor

$$\nabla_j V_i = \frac{\partial V_i}{\partial Z^j} - \Gamma^m_{ij} V_m$$

Index Algebra

1. The Free Index. Indices appearing once on one side must appear once on ther other side.
2. Indices appearing twice on one side must appear in both upper and lower positions. Appearing on both sides actually means that the index on the left and the index on the right are independent of each other (this should be avoided).
3. **Moving from left to right:** An element/tensor (i.e., a factor in multiplication) transfers from one side to the other (i.e., analogous to multiplying by its inverse) by introducing a free index and switching positions yielding (that is, $J^k_b \to J^b_m$ or $Z_{im} \to Z^{mj}$ because $J^k_b J^b_m = \delta^k_m$ and $Z_{im} Z^{mj} = \delta^j_i$)[1]. For example,

$$J^k_b = \delta^c_b \implies J^k_b J^b_m = \delta^c_b J^b_m \implies \delta^k_m = \delta^c_b J^b_m \implies \delta^k_m = J^c_m \tag{1}$$
$$Z_{im} = J^c_m \implies Z_{im} Z^{mj} = J^c_m Z^{mj} \implies \delta^j_i = J^c_m Z^{mj} \tag{2}$$

(1) moves the δ from one side to the other side by removing b and introducing m (both free indices). (2) moves Z to the right (leaving the δ behind) by removing i and introduction j both of which are free indices.

Delta symbols

This symbol determines/filters two permuations that are in sync, out of sync, or neither.

$$e^{ijk} = e_{ijk} = \begin{cases} 1 & ijk \text{ positive permutation} \\ -1 & ijk \text{ negative permuation} \\ 0 & \text{otherwise} \end{cases}$$
$$\delta^{ijk}_{rst} = e_{rst} e^{ijk}$$

Thus, δ^{ijk}_{rst} is positive when ijk and rst have the same permutation sign, negative when they have different permutation signs and zero otherwise. Also, the numbers ijk and rst must come from the same set albeit not the same order (e.g., $\exists i \in rst$ as well as j and k)

Interesting identities

$$\begin{aligned} \delta^{ij}_{rs} &= \delta^i_r \delta^j_s - \delta^j_r \delta^i_s \\ &= \begin{vmatrix} \delta^i_r & \delta^i_s \\ \delta^j_r & \delta^j_s \end{vmatrix} \end{aligned}$$

[1] An expression that results in a Kronecker Delta is being considered a quotient expression

Chapter 10 – Surface Tensors

Shift Tensor

The matrix which converts from the ambient basis, $\mathbf{Z}_i$, and the surface basis, S_α.

$$Z^i_\alpha = \mathbf{S}_\alpha \cdot \mathbf{Z}^i$$

for example.

$$\begin{aligned} \text{Let } \mathbf{T} &= T^i \mathbf{Z}_i \\ &= T^\alpha \mathbf{S}_\alpha = T^\alpha Z^i_\alpha \mathbf{Z}_i \\ \therefore T^i &= T^\alpha Z^i_\alpha \end{aligned}$$

Surface covariant metric tensor

Analogous to the covariant metric tensor Z_{ij}.

$$S_{\alpha\beta} = \mathbf{S}_\alpha \cdot \mathbf{S}_\beta$$

Consequently, the covariant metric tensor and the surface covariant metric tensor are related by

$$S_{\alpha\beta} = \mathbf{S}_\alpha \cdot \mathbf{S}_\beta = \mathbf{Z}_i Z^i_\alpha \ \cdot \ \mathbf{Z}_j Z^i_\beta = Z_{ij} Z^i_\alpha Z^j_\beta$$

The last term is all matrix multiplication starting with the metric tensor that twice the surface metric tensor.

Surface Contravariant Metric Tensor.

This is defined as the matrix inverse that provides

$$S^{\alpha\beta} S_{\beta\gamma} = \delta^\beta_\gamma$$

Note: Z_{ij}, Z^{ij} would be a 3 dimensional square matrix, $S_{\alpha\beta}, S^{\alpha\beta}$ is a two dimensional matrix, Z^i_α is 2×3 matrix and Z^α_i is a 3×2 matrix.

Surface versions of Levi-Civita symbols

Define S (not a vector and no subscripts) as anlogous ot Z (not a vector and no subscripts) as

$$S = |S_{..}|$$

We define the **area element** as $\sqrt{S}$.

Define the permutation operator as

$$e_{12} = -e_{21} = 1 \text{ zero otherwise}$$

then

$$\begin{aligned} \varepsilon_{\alpha\beta} &= \sqrt{S}\, e_{\alpha\beta} \\ \varepsilon^{\alpha\beta} &= \frac{e^{\alpha\beta}}{\sqrt{S}} \end{aligned}$$

AI Questions

Help me understand more about Christoffel Symbols.

Help me understand more about Christoffel Symbols.

The ***Christoffel symbols*** look mysterious at first, but they're really just bookkeeping for how your coordinate system bends and twists.

Let's build intuition step by step.

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1. **What problem are they solving?**

   In flat $\mathbb{R}^n$, derivatives are easy:

   $$\partial_i V^j$$

   works fine because the coordinate axes don't change.

   But on a ***curved manifold*** (or even curvilinear coordinates like polar coordinates), the ***basis vectors themselves vary with position***.

   So when you differentiate a vector field, two things change:

   1. the components V^j
   2. the basis vectors e_j

   Christoffel symbols correct for (2).

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2. **Definition**

   The Christoffel symbols Γ^k_{ij} are defined by:

   $$\nabla_i e_j = \Gamma^k_{ij} e_k.$$

   Interpretation:

   - Take the basis vector e_j
   - Differentiate it in direction x^i
   - The result is another vector, expressed as a linear combination of the basis

   So Γ^k_{ij} tells you:

   how the jth basis vector rotates toward the kth direction when moving in the ith direction.

   —

3. **Covariant derivative of a vector**

   Using this, the correct derivative of a vector field is:

   $$\nabla_i V^k = \partial_i V^k + \Gamma^k_{ij} V^j.$$

   Two parts:

   - $\partial_i V^k$: change in components
   - $\Gamma^k_{ij} V^j$: correction for moving basis

—

4. Why they are NOT tensors

Christoffel symbols are ***not tensors*** because:

- they depend on the coordinate system
- you can make them vanish at a point by choosing normal coordinates

Intuition: They describe the coordinate system as much as the geometry.

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5. Formula in terms of the metric

If you have a Riemannian metric g_{ij}, then:

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell} \left(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij} \right).$$

This shows:

- Christoffel symbols come entirely from how the ***metric changes***
- No curvature yet — just first derivatives of the metric

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6. Geometric intuition

Think of walking on Earth:

- Moving north changes what you mean by "east"
- So even if a vector "looks constant," it actually rotates

Christoffel symbols measure that rotation.

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7. Simple example: polar coordinates

In (r, θ), the metric is:

$$ds^2 = dr^2 + r^2 d\theta^2.$$

Some Christoffel symbols are:

$$\Gamma^r_{\theta\theta} = -r, \quad \Gamma^\theta_{r\theta} = \frac{1}{r}.$$

Meaning:

- moving in θ changes the radial direction
- the angular basis vector shrinks/expands with r

—

8. Connection to curvature

Christoffel symbols themselves are not curvature, but:

curvature = derivatives of Christoffel symbols + quadratic terms

So they are like the "first layer" of geometry.

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9. Connection to your earlier question (normal derivatives)

When you computed:

$$\nabla_i \nabla_j u$$

you were using Christoffel symbols implicitly:

$$\nabla_i \nabla_j u = \partial_i \partial_j u - \Gamma^k_{ij} \partial_k u.$$

That extra term is exactly the same phenomenon:

derivatives must account for changing geometry.

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10. Mental model (most important)

A very useful way to think of Christoffel symbols:

They measure how much your coordinate grid is "curving" or "tilting" as you move.

- If coordinates are straight (Cartesian): $\Gamma = 0$
- If coordinates bend (polar, sphere, manifold): $\Gamma \neq 0$

—

Help me understand the Reimann-Christoffel tensor.

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The ***Riemann–Christoffel tensor*** (usually just called the ***Riemann curvature tensor***) is what finally measures **true curvature**, not just coordinate effects. Let's build it from the ground up in the clearest way possible.

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1. **What is it trying to measure?**
   Christoffel symbols tell you how basis vectors change:

$$\nabla_i e_j = \Gamma^k_{ij} e_k.$$

   But Christoffel symbols can vanish at a point just by changing coordinates — so they ***don't measure real curvature***.
   The Riemann tensor answers a deeper question:
   If you move a vector around a tiny loop, does it come back unchanged?
   - Flat space → it comes back the same
   - Curved space → it comes back rotated

   That rotation is exactly what the Riemann tensor measures.
   —

2. **Definition using covariant derivatives**
   The Riemann tensor is defined by

$$(\nabla_i \nabla_j - \nabla_j \nabla_i) V^k = R^k{}_{\ell ij} V^\ell.$$

   So it measures the ***failure of second covariant derivatives to commute***.
   —

3. **Concrete formula**
   In coordinates, the Riemann–Christoffel tensor is

$$R^k{}_{\ell ij} = \partial_i \Gamma^k_{j\ell} - \partial_j \Gamma^k_{i\ell} + \Gamma^k_{im}\Gamma^m_{j\ell} - \Gamma^k_{jm}\Gamma^m_{i\ell}.$$

   —

4. **What each term means**
   Think of the formula in two parts:
   - (A) Derivatives of Christoffel symbols

$$\partial_i \Gamma^k_{j\ell} - \partial_j \Gamma^k_{i\ell}$$

     This measures whether the "rotation of the basis" changes from point to point.
   - (B) Quadratic correction terms

$$\Gamma^k_{im}\Gamma^m_{j\ell} - \Gamma^k_{jm}\Gamma^m_{i\ell}$$

     These appear because the basis is already rotating while you move.

   —

5. Why this measures curvature

Imagine moving a vector in two steps:

1. Move in direction x^i, then in direction x^j
2. Move in direction x^j, then in direction x^i

In flat space → same result In curved space → not the same

The difference equals

$$R^k{}_{\ell ij} V^\ell.$$

So the Riemann tensor measures how much the vector ***twists due to curvature***.

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6. Geometric intuition

A very concrete picture:

Walk on a sphere

Start at the equator:

- Walk north
- Walk east
- Walk south

Your direction changes even though you never turned.

That change in direction = curvature = Riemann tensor.

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7. What the indices mean

$$R^k{}_{\ell ij}$$

Each index has a meaning:

- i, j: directions you move in
- ℓ: direction of the vector you are transporting
- k: direction it ends up pointing

So the tensor answers:

If I transport a vector in direction ℓ around the i-j plane, how much does it rotate toward direction k?

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8. Why it's a true tensor

Unlike Christoffel symbols:

- You cannot make the Riemann tensor vanish by changing coordinates (unless the space is actually flat).
- If $R^k{}_{\ell ij} = 0$ everywhere → the space is flat.

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9. Important special cases

In 2D surfaces

The Riemann tensor contains only ***one independent number***:

$$K = \text{Gaussian curvature}.$$

So for surfaces:

- sphere $\to K > 0$
- plane $\to K = 0$
- saddle $\to K < 0$

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10. Most useful mental picture

Think of the hierarchy:

Object	What it measures
Metric g_{ij}	distances
Christoffel symbols Γ^k_{ij}	how coordinates bend
Riemann tensor $R^k{}_{\ell ij}$	real curvature

—